Clinical Practice Guideline

Gastrostomy Tubes

Reference #: CPG031

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to the management and care of Percutaneous Endoscopic Gastrostomy (PEG) and Radiologically Inserted Gastrostomy (RIG) tubes. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Table of Contents

1. Principles

Observe the five moments for hand hygiene. Follow aseptic technique and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Manual- Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries

Patient identification must be checked when matching patient's identity to care, therapy and services.

It is a legal requirement that all disciplines that prescribe or administer medications comply with the Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016, Pharmacy Act 2010, and Hospital Guidelines. It is a legislative requirement that a valid prescription must exist before Schedule 4 and Schedule 8 medications are administered.

Only valid prescriptions will be utilised by any nursing staff to administer medication except for verbal orders and nurse-initiated medications- Structured Administration and Supply Arrangements (SASA).

SCGOPHCG recognises the role of carers and shall establish and maintain mechanisms for the involvement of carers in the provision of services

Gastrostomy tubes are a means of providing long term (>4-6 wks.) enteral nutrition. They are inserted through the abdominal wall into the stomach so that liquid nutrition, fluids, and medications can be given.

2. General Information and Care

2.1 Medical Requirements

Informed written consent as per North Metropolitan Health Service Consent to Treatment Policy, and in compliance with the WA Health Consent to Treatment Policy (OD0657/16 v 2.0).

Ensure patient has peripheral vascular access.

Medical Officer to review if patient is prescribed anticoagulation therapy. [1,2 20]

Monitor bloods for potential refeeding syndrome post commencement of feeds.

Review procedural instructions for specific post insertion orders.

Patient may require prophylactic antibiotics.[1,2,7]

2.2 Safety Information

Gastrostomy Contraindications [1, 3,4,5 20]

- Large volume ascites / hepatomegaly / portal hypertension
- Uncorrected coagulopathy, coagulation disorders, gastrointestinal bleeding
- Intraabdominal perforation, bowel ischaemia, intestinal obstruction
- Haemodynamic compromise / sepsis
- Peritonitis / peritoneal dialysis, peritoneal carcinomatosis
- Previous gastric surgery
- Dementia
- Anorexia nervosa
- Severe obesity

Potential Post Procedure Complications[1,3,4,5,20]

- Endoscopic associated complications, over sedation, aspiration, bleeding
- Piercing of colon or small bowel, gastro colic or colo-cutaneous fistula formation

- Inadvertent removal with perforation
- Gastric ulceration / migration with gastric outlet obstruction
- Peri-stomal infection
- Buried bumper syndrome (inner bumper of tube is epithelialized over by the gastric wall requiring surgical removal). [3,6]
- Pneumoperitoneum
- Tube blockage, tube porosity / fracture and subsequent leak, cellulitis, hyper granulation.

Table 1: Types of Gastrostomy Tube

Name	Details	Place of Insertion
PEG Percutaneous Endoscopic Gastrostomy	Non ballooned with internal bumper and tube clamp (Figure 1)	G75 Endoscopy Unit
RIG Radiologically Inserted Percutaneous Gastrostomy	Anchored by a balloon with inflation port and no clamp. 2-3 anchor sutures around site. (Figure 2)	Radiology Department
Surgical Gastrostomy	Gastrostomy tube inserted during elective surgical procedure	Operating theatre

Refer to post procedure instructions for individualised care.
Tube size ranges from 9-28fr
An external fixation plate or flange (figure 1) keeps tube from rubbing around the insertion site, protects the skin from damage and prevents the tube being drawn into the gut by peristalsis.

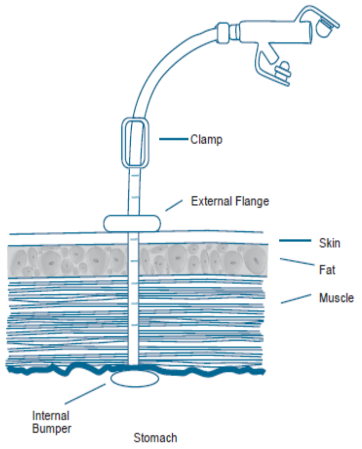


Figure 1: PEG: Non-ballooned tube

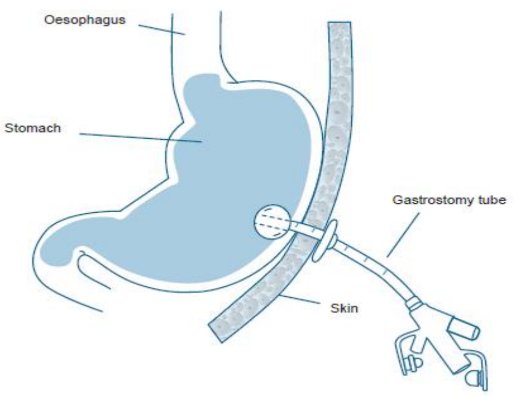


Figure 2: RIG: Ballooned tube

Tubes may be changed to a low-profile device which sits at skin level and requires a detachable extension tube for feeding or medication administration (for specific instructions / information on use of low-profile devices and extension sets see instructions for use available from radiology or G75 **Low Profile (MIC- KEY*)** has an internal balloon to secure the device.

- **Low profile obturate gastrostomy** is anchored by a mushroom shaped internal flange that is stretched into shape for insertion. (Figure 3)

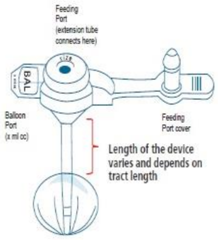


Figure 3:
MIC-KEY* Low Profile Device With ballooon

Obturate Low Profile Device

Continuous feed Extension Set

Bolus Feed Extension Set

2.3 Nursing Management

Pre-Procedure

- Complete appropriate checklist.
- Patients must be fasted 6 hours solids and 2 hours clear liquids, longer if impaired gut motility [3]
- Administer medications as NPG 55 Preoperative care
- Patients being prepared for insertion of PEG / RIG are to shower / bathe using Chlorhexidine 4% wash on the morning of the procedure.
- Obtain any specialist nursing advice from the Clinical Nurse Consultant Endoscopy Unit or Radiology Day Unit Special Procedures.
- Contact Dietitian to inform of tube insertion. Feeding via tube may commence 3-4 hours post insertion or as per post-procedural instructions; oral intake if appropriate may also commence at this time.[3-5]

Immediate Care post insertion

- Record tube type, brand, size and the baseline length of tube. For tubes with measurements on them, record the figure at skin level. For tubes with no measurements, mark at skin level with permanent marker and record length of the tube from skin to base of the feeding adaptor.[2,5]
- Post procedure vital signs and escalation as per Observation Response Chart (ORC). [7,8,9,20]
- Complete abdominal assessment for sign of peritonism including abdominal pain, guarding, distension, rigidity or bloating, observe for moderate to large amounts of fresh bleeding, haematoma, redness, excessive swelling, discharge or leak of gastric contents **1hrly for 4hrs, 4hrly for 24hrs, then once per shift** - report to medical officer. [4,5,10]
- Observe for increased pain post feed, diarrhoea, nausea, or vomiting, decreased urine output, absence of flatus, bowel sounds or inability to open bowels.[7,8,9]
- Assess pain score and optimise pain management, patients requiring escalation of analgesia require a medical review.[7,8,9 20] Pain can continue for up to 7 days.[7,9]
- Leave dressing intact for 24hrs unless copious discharge, patient complains of discomfort or excessive oedema is present. If these occur change dressing as below.[11]

Ongoing Care after 24hrs: (continue to observe for potential complications above).

Insertion Site / Dressing

- Post insertion and for 14 days dress insertion site daily or if exudate is present using aseptic technique clean with sterile normal saline and use sterile gauze or foam keyhole dressing. Place dressing on top of external flange around tube, cover with a non-occlusive dressing, aim to keep wound dry. [2,3,10,11,12]

- Remove crusts or exudate from close to the tube and under the external flange with sterile normal saline and a sterile cotton-tipped applicator. Dry completely. [5,7,8,9]
- When healed (approx. 14 days) cleanse daily with soap and water and dry thoroughly, no dressing is required (patients may shower). [2,5,11,12]
- Inspect site at least daily for infection, erythema, exudate, leakage, induration, hyper-granulation and document. [5,11,12]
- Avoid use of creams or powders round the stoma.[5,11,12]
- Notify shift coordinator, Clinical Nurse Specialist/ Clinical Nurse Manager, and medical officer of any changes (e.g., If gastric contents are leaking around site and causing skin excoriation).
- Contact Clinical Nurse Manager (G75) if problems persist with stoma site.
- Check external flange with patient in sitting and reclined position, to ensure not too tight or too loose, correct position is 2-5mm from flange to skin level to avoid tissue compression between internal and external flange / balloon. [1, 3, 4, 6]

Rotation of Tube / Balloon Volume

PEG:

- Commence rotation of tube 7 days post insertion, clean insertion site if required prior to rotation.[5,7,10,11,12]
- Rotate tube 360 degrees daily.[6,7,12] Push tube forward into the wound 1-2cm and rotate to ensure internal bumper does not become buried into gastric mucosa, then return to original position. Record in inpatient record and nursing care plan..[3,6,7,8,11]

RIG

- Newly inserted RIGs have anchor sutures, 2-3 button-like attachments around the site. These are usually removed after 7 days. [3,11,12] Lift the external flange and cut the suture underneath. Sutures are absorbable so may fall out on their own. Radiology preadmission clinic nurse will follow up with patient via phone call if discharged or with inpatient ward to ensure sutures removed. [20]Do not rotate tube until sutures removed. Then rotate tube as for PEG.
- 4-6 weeks post insertion - internal balloon volume should be checked weekly. Remove water with syringe and compare amount removed to recommended amount. / previous amount instilled. Refill the balloon up to recommended amount with water. Do not use air or saline to fill the balloon, do not overinflate.[10,12, 21]

Tube Position

- Check tube length (in inpatient record / on nursing care plan). If measurement deviates from initial recording this may indicate tube migration, report to medical officer, do not use until reviewed.[5,7,8,9]
- Maintain tube position and avoid traction on tube by using suitable catheter fixation device, alternate position to avoid skin irritation. [5,8]

Gastrostomy Tube

- Tube must be flushed daily with tap water to **prevent blockage if not otherwise in use**. Sterile water should be used for immunocompromised and critically ill patients. [2,4,5,7,13,14, 20]
- If clamp present leave closed when not in use, change position along device regularly, ensure clamp open prior to use. [7,9]
- Never flush or administer feed via the inflation port. [9]
- If 'Y' port present and becomes damaged this can be replaced.

Mouth Care

- A minimum of twice daily oral hygiene should include gentle brushing of teeth, gums, tongue and palate with soft toothbrush and mild toothpaste. Suction can be used for patients with oropharyngeal dysphagia. This includes patients who are nil by mouth (NBM) or edentate. Patients who are NBM are at high risk of overgrowth of the microbiota.[7,15]

Selection of Enteral Dispensers / Syringes

- Enteral dispensers / syringes and feeding systems must not contain ports that can be connected to intravenous devices.

Methods for Enteral Nutrition Delivery

- Feeding may be administered by continuous, intermittent or bolus methods.

Bolus is the administration of enteral feeding via

- Syringe / enteral dispenser (gravity administered) for a period of 5-10 minutes every few hours
- Feeding bag (gravity) administered usually over 15-60 minutes (e.g. every 4 hrs.)
- Enteral feeding pump, which delivers feed at a prescribed rate and reduces the risk of tube blockage.[16]

Continuous Feeding

- Feeding is delivered continuously over any specified period of time over a 24-hour period via an enteral feeding pump

Cyclic /Intermittent Feeding

- Enteral nutrition may be ceased for a 4-16 hour period during the day or night. Can be delivered via pump or gravity drip

3. Administering Medications

3.1 Medical Requirements

A valid prescription must be present on the WA Medication Chart

Refer to Formulary One, or ward pharmacist for alternative routes, crushing instructions and medication preparations for administration via enteral tubes. [10]

3.2 Safety Information

Inform the medical officer of any signs indicating aspiration, adverse drug reactions, tube migration or dislocation.

Nil via gastrostomy tube 3-4 hours post insertion, unless otherwise instructed.[3,3,7,13,16]

Position the patient at 30–45-degree angle (head/upper body elevated) during, and 30-60 minutes after, administration to reduce risk of aspiration. If contraindicated, consult with medical officer regarding an alternative.[7,6,13,16]

Oral medication route should be used if patient has a safe swallow.

Consult with pharmacist regarding appropriate preparations for medication administration and timing of administration (with respect to feeding) for <u>all</u> medications.[8,13,16]

Preparations should be liquid, effervescent or dispersible if possible. Alternative routes such as parenteral, transdermal, sublingual, inhaled or rectal should be considered.[3,46,8,12,16,20] bulking agents (psyllium) and resins (e.g., cholestyramine) should not be given via enteral tubes. [6]

Many liquid medication preparations contain sorbitol, which can interact with enteral formulas and block the tube [3,17]

Medications to be taken on an empty stomach should be administered 30 mins before or 2 hours after feed. [3,16]

Medications should preferably be administered using gravity flow with the enteral dispenser barrel. It may be necessary to use the enteral dispenser and plunger with gentle pressure when administering medications.

Administer each medication individually (not mixed with other medications or formula) and flush with 30mls water before, between each preparation and after administration to avoid clogging. [3,7,16]

Consider adapting flushing protocols in patients with restricted fluid intake, e.g., 10ml every 6hrs with continuous infusions; and 5ml before and 10ml after medications or interrupting starting feeds[2,4,6,8,12,13].

3.3 Nursing Management

Check tube placement / measurements prior to administration (see baseline measure in inpatient record / nursing care plan). Withhold administration if migration or dislocation has occurred and consult with medical officer

Use only enteral dispenser / syringes to measure and administer medications via enteral tubes. [3,8]

Enteral feeding should be delayed 1 – 2 hours where a drug interaction is known (e.g., Warfarin, Phenytoin, Ciprofloxacin and Sinemet™).[8]

3.4 Procedural Information

Requirements

- Prescribed medication (s)
- 60ml oral/enteral dispenser (single use only)
- Medicine cup (one per medication)
- Cup of warm tap water (sterile water if immunocompromised)
- Pill crusher
- Personal protective equipment

Procedure

1. Perform hand hygiene.
2. Prepare prescribed medications separately, one medication per medication cup. [7]
3. If required crush each medication finely and mix with at least 30mls warm water. [13]
4. Perform hand hygiene and apply PPE
5. Position the patient at 30-45 degree angle (head/upper body elevated, unless contraindicated) during, and at least 30 minutes after administration to reduce risk of aspiration. [6,8,16]
6. Stop feed (if running).
7. Flush with water 30 ml prior to administering medication. [2,4,7]
8. Administer first medication with a clean enteral dispenser, preferably using gravity flow.
9. Clamp or kink tube between administrations
10. Flush with 30ml of water between each medication.[2,7,13]
11. Complete administration with final flush 30ml of water. [7,13]
12. Clamp (if present) tube for 1 hour if no feed is running or restart feed. [8]
13. Discard enteral dispenser (single use).
14. Remove PPE and perform hand hygiene.

4. Feeding

4.1 Medical Requirements

The dietitian in consultation with medical team prescribes volume of feed. Fluid assessment is to be carried out to determine the need for additional fluid requirements.

Consider refeeding syndrome; check potassium, phosphate and magnesium levels prior to commencing enteral feeding and replace if necessary. Continue to monitor daily until corrected.

Newly inserted tubes refer to post procedure instructions, feeding with formula can usually commence 2-4 hrs. post procedure. [3, 7,10,]

4.2 Safety Information

Refeeding Syndrome: If the patient has been fasted or consuming <25% of their meals for 7-10 days or more, they may be at risk of refeeding syndrome, a set of metabolic disturbances which can arise when a patient is fed above their current threshold level (orally, enteral, or parenterally) after a period of starvation. Refer to the Oral and Enteral Refeeding Syndrome Guideline for more information

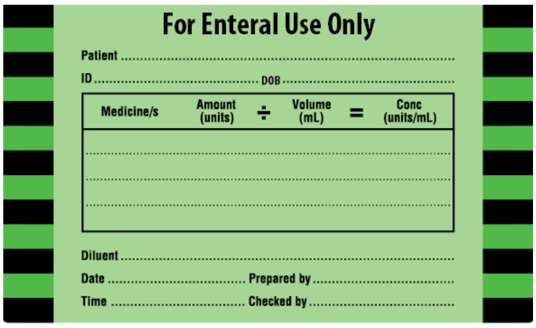
Position the patient at 30–45-degree angle (head/upper body elevated) during, and 30-60 minutes after, administration to reduce risk of aspiration. If contraindicated, consult with medical officer regarding an alternative.[7,13]

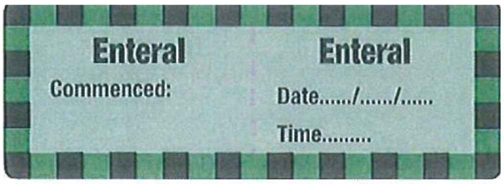
Notify medical officer of:

- Respiratory difficulties or vomiting/regurgitation
- Signs of feed intolerance such as distension, nausea, vomiting, constipation, reflux, discomfort, or diarrhoea (also notify dietitian)

Check tube placement / measurements prior to administration of feeds. Withhold administration if migration or dislocation has occurred and consult with medical officer. [8,18]

The route of administration must be identified on all administration lines and include the date and time that the line is commenced.





Pre-filled Ready to Hang (RTH): Feeds are a closed system and packs may be hung at room temperature for up to 24 hours or as per manufacturer's instructions.[13] Change administration set each time a new Prefilled RTH Feed is used up to maximum of 24hrs. [14]
Bottle, tetra pack or can pre-packaged feeds: Decant a maximum of 8hrs of feed into administration set / bag. Only top up when feed is almost empty. [13,14]
When a patient is required to fast prior to a procedure enteral feeding is to be discontinued.
Kangaroo ePump™ Set must be changed every 24 hours.[14] gravity feed sets, and oral / enteral dispensers (syringes) are single use. [13,14]

When a patient is required to fast prior for a procedure, enteral feeding must be discontinued.

4.3 Nursing Management

- Feeding regimes will be documented by the dietitian on either: Bolus Enteral Nutrition Care Plan, Continuous Enteral Nutrition Care Plan or Transitional Enteral Nutrition Care Plan

Feeding intolerance

- Monitor the patient at least daily for signs and symptoms of intolerance to enteral feeding. Absence of bowel sounds / passing stool, discomfort or pain, nausea or vomiting, diarrhoea or abdominal distension may be signs of feed intolerance.[7,13]
- If intolerance is suspected discuss with dietitian and medical staff for consideration of prokinetics, post pyloric feeding or parenteral nutrition.

Cessation or interruption of feeding should be avoided to minimise the risk of ileus and to avoid inadequate nutrition. Minimise time patient is nil orally prior to surgery / diagnostic tests.

Notify Dietitian if there is an unscheduled stop in feeds.

If the feeding regimen is altered and the patient is receiving insulin therapy, insulin requirements may need to be adjusted, liaise with Medical Officer to determine ongoing management.

- Feeding may commence 3-4 hours post insertion. [2,18]
- Document volume of feed / fluid administered on fluid balance chart.

4.4 Procedural Information

Requirements

BOLUS FEEDING

- Prescribed Enteral Formula (check integrity and expiry)
    - o 60 ml oral enteral dispenser (Enteral dispenser gravity feeds) **OR**
    - o Gravity feeding set (gravity feeds) **OR**
    - o Epump Set **OR**
    - o Epump ENPlus 3 in 1 Set and ready to hang feed. (Continuous / Intermittent / Bolus Feed with pump)
- IV Pole
- Plastic cup
- Cup of tap water (sterile water if immunocompromised) [4,7,8,13,14,17,18]
- Kidney dish
- Personal Protective Equipment
- Enteral Labels

Bolus Feeding:
<u>Enteral Dispenser Gravity Feed (syringe) [10]</u>

1. Perform hand hygiene.
2. Clean work surface with pre-diluted detergent, or a detergent wipe and allow to air dry.
3. Gather equipment.
4. Perform hand hygiene and apply PPE.
5. Position patient with head of bed elevated to at least 30 degrees unless contraindicated
6. Verify correct tube placement by checking external tube length [2,7]

7. Flush with 30mls water. [10]
8. Remove plunger of enteral dispenser and connect dispenser to tube. Gently kink tube, this is to prevent air being drawn into the stomach.
9. Pour required amount of feed in 50ml increments into dispenser barrel. Release clamp if present
10. Raise or lower enteral dispenser height to allow gravity to control the administration rate. If initially difficult to commence feed, insert plunger into barrel of dispenser and apply gently pressure to commence. Remove plunger and raise arm.
11. Complete feed by administering prescribed amount of water.
12. Discard any unused feed appropriately– DO NOT refrigerate or keep.
13. Discard enteral dispenser (single use). Remove PPE and perform hand hygiene.
14. Document quantity of fluid administered on the Fluid Balance Chart (MR904)

Enteral Administration Gravity Feeding Set

1. Perform hand hygiene.
2. Clean work surface with pre-diluted detergent, or a detergent wipe and allow to air dry.
3. Gather equipment.
4. Perform hand hygiene and apply PPE.
5. Position patient with head of bed elevated to at least 30 degrees unless contraindicated
6. Verify correct tube placement by checking external tube length
7. Flush with 30mls tap water (sterile water if patient is immunocompromised). [14]
8. Discard enteral dispenser.
9. Prepare Gravity Feeding set connect and prime with enteral formula and attach to enteral feeding tube. Do not fill with > 8 hrs. volume of formula.
10. Complete Enteral Label and affix to enteral administration set
11. Administer prescribed volume of feed at rate ordered.
12. Complete feed by administering prescribed amount of water.
13. At completion of feed disconnect set tube, and discard
14. Discard any unused feed – do not refrigerate or keep. Remove PPE and perform hand hygiene.
15. Document the volume of fluid administered on the Fluid Balance Chart.

Continuous / Intermittent / Bolus Feed with Kangaroo EPump)-

1. Perform hand hygiene.
2. Clean work surface with pre-diluted detergent, or a detergent wipe and allow to air dry.
3. Gather equipment.
4. Perform hand hygiene and apply PPE.
5. Position patient with head of bed elevated to at least 30 degrees unless contraindicated
6. Verify correct tube placement by checking external tube length
7. Flush with 30mls tap water (sterile water if immunocompromised)[9, 10]
8. Remove PPE and perform hand hygiene
9. **EITHER:**
   a) Decant up to 8 hrs. of enteral feed into Kangaroo Epump Set, **OR**
   b) Connect Kangaroo Epump ENPlus 3 in 1 Set to ready to hang (RTH) feed.
10. Complete and affix Enteral Label to load and prime the Kangaroo ePump™:
11. To load and prime the Kangaroo ePump™:
    a) Locate power button on front of control panel, press to turn on.
    b) Confirm if you want to keep or clear any prior pump settings

c) Open the blue door to load pump set, insert enteral administration set and close the blue door. Display will read SET LOADED.
d) Press 'PRIME', followed by 'AUTO PRIME' to automatically prime the administration set. Alternatively, to manually prime the administration set, press the 'HOLD TO PRIME'. Press 'DONE' when complete.

12. . Perform hand hygiene and apply PPE.
13. Connect the administration set to enteral feeding tube.
14. 10. Remove PPE and perform hand hygiene.
15. Program the prescribed rate (ml/hr.):
    a) Press 'ADJUST FEED' to program the hourly feed rate and volume to be delivered.
    b) Press 'ENTER' to confirm feed rate (ml/hr.) and volume to be delivered.
    c) Press 'DONE' to return to main screen and then press 'RUN' to commence the feeding regime.
16. **Flush tube with 30mls of water, every 4 hours and at the completion of the feed.** [4,7,8,10,18]

At completion of:

- intermittent / bolus feed, disconnect set from enteral tube and discard,
- continuous feed, enteral feed administration sets must be changed every 24hours.

17. Discard any unused feed – do not refrigerate or keep.
18. Document quantity of fluid administered on the Fluid Balance Chart

5. Blocked Tubes

5.1 Medical Requirements

Inform medical officer if attempts to unblock are unsuccessful.
A valid prescription (using WA Medication Chart) for pancreatic enzymes is required.

5.2 Safety Information

Do not use excessive force when attempting to unblock tube. Do not use syringe smaller than 10ml. [12,17]

5.3 Nursing Management

Most common causes of tube blockages include: [4,7,13,17]

- Failure to flush the tube at regular intervals
- Administration of partially crushed medications
- Feed / drug interactions
- Kinked tube
- Siphoned gastric fluid from not using the clamp
- Material fatigue associated with aging or mishandling the tube

Warm tap water (sterile water if immunocompromised or post pyloric tube) should be used as first line to unblock tubes.[4,7,8,13 20]
Pancreatic enzyme mixture can be used to unblock tubes. Consult with medical staff for prescription.[2, 9]

Pancreatic Enzyme mixture = 10ml warm tap water, 1 capsule sodium bicarbonate (or one 10ml vial of 8.4% sodium bicarbonate or ¼ teaspoon bicarbonate of soda), 1 capsule Creon Forte 25,000 units. Crush Creon well, mix ingredients together.[4,6,7,8,17,20]

If attempts to unblock tube are unsuccessful, please notify shift coordinator, Clinical Nurse Specialist, medical officer and liaise with Clinical Nurse Manager Endoscopy G75.

5.4 Procedural Information

Requirements

- 60ml enteral dispenser (smaller syringe can be used with caution) [4,13]
- Plastic cup
- Absorbent sheet
- Kidney dish
- Personal Protective Equipment
- Warm tap water tap (sterile water if patient is immunocompromised). If ineffective pancreatic enzyme mixture. [17]

Procedure

1. Perform hand hygiene.
2. Clean work surface with pre-diluted detergent, or a detergent wipe and allow to air dry.
3. Gather equipment.
4. Perform hand hygiene and apply PPE.
5. Visually inspect tube for occlusions, check clamp and connectors.
6. Ensure tube moves freely in and out of abdomen to exclude “buried bumper” as cause of blockage.[17]
7. Gently rolling the tube between thumb and forefinger may help to break down the blockage in the tube17
8. Using an enteral dispenser try to aspirate tube before attempting to unblock.
9. If unable to aspirate, attempt to flush the tube with 10mLs warm water using the enteral dispenser, applying a continuous pulsating push / pull motion (this may take several minutes).[8,13,17]
10. If able to insert some solution but unable to flush fully leave water / solution insitu for 30 minutes and then repeat.[13,17] (Or if unsuccessful pancreatic enzyme mixture).[17]
11. Aspirate as much of tube content out of tube as possible.[17]
12. If attempt to unblock the tube is unsuccessful, notify Shift Coordinator and Medical Staff to determine ongoing management.
13. If blockage is cleared, flush the tube with 30mls of water.
14. Remove PPE and perform hand hygiene.
15. Document in the patient progress notes.

6. Accidental Removal / Dislodgement of Tube

6.1 Medical Requirements

Consult with the medical officer whenever accidental removal occurs.

OPH only: If tube has been completely removed, temporary use of a Foley’s catheter can be used to maintain the tract. This is not recommended within the first 2-4 weeks of tube placement due to the risk of intraperitoneal insertion.

- Check replacement with pH confirmation of gastric contents (pH 5 or less), irrigation of tube with 30-50mls sterile water without resistance or leakage around the stoma, assessment of external length of tube.
- If there is doubt regarding correct placement endoscopic or radiologic confirmation may be required. Foley catheter must not be used for medication or feed. [3,6,10]

6.2 Safety Information

Do not attempt to insert a replacement or temporary tube on the ward.

6.3 Nursing Management

- Accidental removal should be reported to the medical team (on call doctor after hours) as soon as this is discovered. During working hours, also contact the Endoscopy Unit.
- Arrange gastrostomy tube replacement with the Endoscopy Unit or the Radiology Department, as per the medical officer's advice as soon as practicable
- Planned removal of a Gastrostomy tube is only performed by the Endoscopy Unit (G75) or in the Radiology Department.

7 Discharge Planning

7.1 Medical Requirements

Prescriptions for appropriate medications with relevant administration instructions are provided on discharge.

E-referral to the Gastroenterology Service -G75 PEG clinic is required for booking of follow-up appointments.

7.2 Safety Information

Ensure competence of patient / family / carers or the home care service in gastrostomy tube maintenance and feeding[7,8,10,13,18]

7.3 Nursing Management

Patient / Carer Education:
Dietitian will provide the patient / carer with:

- **Home Enteral Feeding Book** (for RIG or PEG): which contains step by step instructions
- Dietitian and Nursing Staff are to educate patient / carer / care facility regarding ongoing management of enteral tube and feeding regime. Dietitian will complete the Enteral Feeding Discharge Summary
- Nursing staff are to assess ability to self-care for all aspects of enteral tube management prior to discharge. and record competency in the Patient / Carer Pre-discharge: Enteral Feeding Competency Checklist for Nursing Staff prior to discharge
- Nursing staff in Radiology will provide RIG Suture Removal Guide Handout for RIG insertions.

Nursing staff to explain and / or demonstrate the following then assess the patient / carer's knowledge and practical ability to complete required care and document the outcome in the patient record:

- Infection prevention including hand hygiene and mouth care [14]
- Unblocking and flushing tube,
- Preparing formula,
- Administering formula by enteral dispenser / gravity / pump as appropriate,
- Medication administration,
- Flushing the tube,
- Checking position (length of tube) [13]
- Correct positioning for feeding,
- Stoma care / dressings / rotation of tube / balloon volume monitoring (G75 CNC may complete),
- Storage of formula,
- What to do if tube is dislodged.

Patient / carer may not be able to perform all aspects of care, individualise as appropriate: refer to community support services / G75 CNM for support.

Discharge advice should also include information on: (see Home Enteral Feeding Book)

- Feeding regime and where to obtain feeds from (dietitian)
- Medication management (pharmacist)
- What to do if there are complications
- Ensuring change of tube occurs when needed
- Obtaining equipment (e.g., IV pole) and supplies (Surgical House).

Patient must be discharged with all required equipment and appropriate follow up appointments to monitor progress and review feeding regime.

Dietitian will provide:

- 7-day supply of enteral feeds
- Educational resources
- Prescription for enteral feeds

Nursing staff to:

- Supply 7-day supply of consumable equipment i.e., enteral dispensers, gravity feeding bags, adaptors as required. Refer patients discharged home to Silver Chain, post-acute care team

Dietitian to refer to rehab in the home (RITH) Dietitian

Provide contact details for Residential Care Line if patient is discharged to a residential aged care facility.

Provide patient with details of timeframe for change of tube and/or gastroenterology G75 outpatient appointment (RIGs at approx. 3-6 months, PEGs up to 2 years or if problems occur.) Note: E-referral to the gastroenterology G75 PEG clinic is required for booking of follow-up appointments.

Document on the Discharge / Transfer List equipment and information provided.

8 Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Carer's Recognition Act 2004 (WA).
- Work Health and Safety Act 2020
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018
- Western Australia: Medicines and Poisons Act 2014, Medicine and Poisons Regulations 2016, Pharmacy Act 2010

Policy:

- WA Health Mandatory Policy Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- SCGOPHCG Patient Identification
- OPH Procedure Matching Policy
- SCGH Procedure Matching Policy
- SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy
- WA Department of Health Mandatory Policy 0131/20 High Risk Medication Policy.
- Australian Commission on Safety and Quality in Health Care. National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines. Sydney: ACSQHC, 2015.
- SCGH Nutrition and Diet therapy -Oral and Enteral Refeeding Syndrome Protocol
- SCGH Dietetics and Nutrition Department Initial Enteral Feeding Regimen Practice Guideline

- SCGH Intensive Care Unit (ICU) Enteral Feeding guideline.

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